Course: Big Data Analytics Laboratory

Course Code: 303105362

Prerequisite: Database Management System, SQL

Rationale : Big data analytics is the often-complex process of examining big data to uncover information such as hidden patterns, correlations, market trends and customer preferences that can help organizations make informed business decisions.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/ Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
0	0	2	0	1	-	-	20	-	30	50

SEE - Semester End Examination, **T** - Theory, **P** - Practical

List of Practical

1.	To understand the overall programming architecture using Map Reduce API.
2.	Write a program of Word Count in Map Reduce over HDFS.
3.	Basic CRUD operations in MongoDB.
4.	Store the basic information about students such as roll no, name, date of birth, and address of student using various collection types such as List, Set and Map.
5.	Basic commands available for the Hadoop Distributed File System.
6.	Basic commands available for the HIVE Query Language.
7.	Basic commands of HBASE Shell.
8.	Creating the HDFS tables and loading them in Hive and learn joining of tables in Hive.

Course Outcome

After Learning the Course the students shall be able to:
1. Understand the Big Data flow.
2. Solve problems using MapReduce technique.
3. Implement single-node/multinode Hadoop cluster.
4. Differentiate between conventional SQL query language and NoSQL.
5. Apply the various technologies and tools associated with Big Data such as HDFS, MapReduce, Pig, Hive, MongoDB.